

EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Software Requirements

- Microsoft Power BI Desktop
 - Microsoft Excel
 - CSV Dataset
 - Windows Operating System
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Algorithm 1

Importing Data into Power BI

Step 1

Open Power BI Desktop

Step 2

Select:

Home → Get Data

Step 3

Choose Excel or CSV

Step 4

Browse dataset

Step 5

Click Load

Step 6

Verify imported data

Procedure

Home Screen

Home

|

|-- Get Data

|-- Excel

|-- CSV

|-- SQL Server

Output

Imported dataset displayed in Data View.

Product Region Sales

Laptop North 50000

Mobile South 35000

Printer East 25000

Algorithm 2

Data Transformation using Power Query

Step 1

Select Transform Data

Step 2

Open Power Query Editor

Step 3

Remove duplicates

Step 4

Handle null values

Step 5

Rename columns

Step 6

Apply changes

Example Transformation

Before

Product Sales

Laptop 50000

Mobile Null

Printer 25000

After

Product Sales

Laptop 50000

Mobile 35000

Printer 25000

Output

Cleaned dataset loaded successfully.

Algorithm 3

Creating a Bar Chart

Step 1

Choose Bar Chart Visualization

Step 2

Drag Product → Axis

Step 3

Drag Sales → Values

Step 4

Format chart

Output Visualization

Product-wise Sales

Sales comparison across products.



Interpretation

Laptop generated highest sales.

Printer generated lowest sales.

Algorithm 4

Creating a Pie Chart

Step 1

Select Pie Chart

Step 2

Drag Product → Legend

Step 3

Drag Sales → Values

Step 4

Apply Data Labels

Output Visualization

Sales Contribution

Percentage contribution of each product.



Interpretation

Laptop contributes the highest share of total sales.

Algorithm 5

Creating a KPI Card

Step 1

Select Card Visualization

Step 2

Drag Sales Field

Step 3

Apply Formatting

DAX Formula

Total Sales

Total Sales = $\sum Sales$

DAX Measure

Total Sales =
SUM(Sales[Sales])

Output

KPI	Value
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Total Sales	225000
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Algorithm 6

Creating Slicers

Step 1

Select Slicer Visualization Step

2

Drag Region Field

Step 3

Enable Single Selection

Step 4

Apply Filters

Output Available

Filters:

- North
- South
- East
- West Dashboard

Behavior When North is
selected:

Product Sales

Laptop 50000

Mobile 40000

Dashboard automatically updates.

Algorithm 7

Creating DAX Calculations

Average Sales

Average Sales =
AVERAGE(Sales[Sales])

Maximum Sales

Maximum Sales =
MAX(Sales[Sales])

Minimum Sales

Minimum Sales =
MIN(Sales[Sales])

Output

Measure	Value
Average Sales	37500
Maximum Sales	50000
Minimum Sales	25000

Algorithm 8

Creating Dashboard

Step 1

Add KPI Card

Step 2

Add Bar Chart

Step 3

Add Pie Chart

Step 4

Add Slicer

Step 5

Arrange Visuals

Step 6

Save Dashboard

Dashboard Layout



Result:

Data visualization using Power BI has been done successfully.